Delta-Hedged Options Strategy

Predicting stock moves around earnings using company financials and machine learning

Utsav Thakur — `utsavthakur`
Vidhi Jain — `vidhi`
Bailey Meche — `baileymeche`
Mohamed Zakaria — `mohzak`
Madhav Fadadu — `madhavf`

FINM Quantitative Trading Project · 2010–2024

Strategy Overview

Goal: Forecast which stocks will move the most after their earnings announcements, using company financial data and a machine learning model. Trade those predictions through options to profit from big price moves regardless of direction.

Collect Data → Align Timelines → Build Features → Select Universe → Train Model → Rank & Signal → Trade Options

58 stocks Dynamic universe, refreshed annually from S&P 500

13 features Company financial changes + price and trading activity signals

15 years Walk-forward training: model only sees past data, never the future

109K predictions Strictly out-of-sample forecasts across 2010–2024

Why This Strategy?

The Opportunity

- When companies report new financial results, stock prices don't adjust instantly
- Different investors interpret the same information at different speeds
- This creates a short window where prices haven't fully "caught up" to the new information

What Research Shows

- "Post-Earnings Drift" - stock prices continue moving in the same direction for weeks after earnings surprises
- Institutional investors reposition gradually over days, not instantly
- The size and speed of the reaction depends on the company's recent history

Our Bet

- A machine learning model can learn to spot which companies will move the most after earnings
- By looking at patterns in company financials + recent trading activity over the past 60 days
- We trade those predictions through options to profit from big moves in either direction

How It Works

1 **New Numbers Come Out**

Companies report quarterly earnings. These financials aren't usable until ~45 days after the quarter ends (filing lag). We never peek ahead.

2 **Investors React Slowly**

Some investors react in hours, others in days or weeks. The speed depends on the company's recent history, how surprising the numbers were, and market conditions.

3 **We Predict the Move**

Our model reads 60 days of financial changes + market signals for every stock. It predicts which stocks will have the biggest price reactions after their next earnings.

The key insight: company financials arrive slowly, but prices move daily - this mismatch creates a window we can trade

The Model: Why LSTM?

Think of it like a golfer adjusting during a round

After every shot, a golfer remembers the useful information - how the wind affected the ball, how the club performed - and forgets the rest. They don't treat every past shot equally; they focus on what matters for the next one.

Our LSTM works the same way. It reads 60 days of data for each stock, decides what's important to remember, what to ignore, and predicts what happens next.

Traditional Approach

- Looks at today's data only, in isolation
- Assumes fixed, linear relationships
- Can't learn from patterns over time

Our LSTM

- Reads 60 days of history for each stock
- Remembers what matters, forgets the noise
- Learns evolving patterns and regime shifts

Data Sources & Universe

Dataset	Source	Size	What It Tells Us
Company Financials	Zacks	405,705 rows	Quarterly earnings, margins, leverage, book value
Stock Prices	PRICES.csv	19.2M rows	Daily prices, volumes, adjusted for splits & dividends
Options	WRDS OptionMetrics	2.5M rows	ATM call option prices, delta, implied volatility

Dynamic Universe Selection

(not a fixed set of stocks - refreshed every year to stay current)

- Start with all S&P 500 members each year
- Rebalance annually on the first trading day after May 15 (2006–2024)
- Apply filters: must have options data, sufficient price history, stable beta, and adequate liquidity
- Exclude Financials and Utilities (distorted accounting ratios)
- Cap at 8 stocks per sector to ensure diversification
- Target: 30 stocks per year, with a turnover buffer to prevent excessive churn

19 annual rebalances · 30 names per year · 58 unique tickers across the full sample

No Peeking: How We Prevent Cheating

The biggest risk in backtesting is accidentally using information that wasn't available yet

A company's quarter ends Dec 31, but the numbers aren't filed until mid-February. We add a 45-day buffer before using any financial data - the model never sees information before investors could have.

Timeline of a Single Trade

Quarter Ends → +45 Days Data Available → Model Predicts → Trade Next Morning → Profit/Loss Realized

Three Anti-Cheating Safeguards

45-day filing lag	We wait 45 days after quarter-end before using any financial data
Walk-forward training	The model is retrained each year using only data from prior years - never the future
Automated validation	Code checks every data point and hard-crashes if any future information leaks in

What the Model Looks At

13 signals, organized in two groups:

Company Financial Changes (8)

Measures how fundamentals are changing quarter-to-quarter - not where they are, but which direction they're moving.

- Profitability change (ROE)
- Operating margin change
- Leverage (debt) change
- Earnings growth (EPS)
- Book value growth
- Plus: acceleration and surprise variants

Market Context (5)

Captures recent price behavior and trading activity - helps the model understand the market environment.

- Recent stock return
- Market sensitivity (beta vs S&P 500)
- Unusual trading volume
- Price reaction speed
- Volatility regime (short vs long term)

All features are standardized across stocks each day so the model compares apples to apples, not absolute size

How the Model Trains

What Goes In / Comes Out

- Input: 60 days of 13 features for each stock
- Output: predicted next-day stock return
- Trained across all 58 stocks simultaneously - learns patterns that work across companies
- 124,824 training sequences total

How We Prevent Overfitting

Train 2006–2009	Predict 2010
Train 2006–2010	Predict 2011
Train 2006–2011	Predict 2012
...	...
Train 2006–2023	Predict 2024

Key Discipline

- The model is retrained fresh each year using only older data - it never sees the future
- 15 out of 15 annual folds trained successfully
- 109,066 predictions produced, all strictly out-of-sample
- This mimics how a real fund would operate: always learning from the past, predicting the future

From Predictions to Trades

Step 1: Rank Stocks

- Each day before an earnings announcement, the model predicts returns for all stocks
- We rank them: #1 = highest predicted move
- Pick the top 3 (K = 3) for trading
- Tie-breaker: prefer the more actively traded stock

Step 2: Earnings Trigger

- We only trade around quarterly earnings - that's when fundamentals matter most
- Prediction is made the day before the announcement
- Entry is the day after (post-announcement, after the initial volatility settles)
- 458 earnings events over 2010–2024

Step 3: Delta-Hedged Options

- For each selected stock, we buy a near-the-money call option (30–45 days to expiry)
- Simultaneously short shares of the stock to neutralize directional risk ("delta hedge")
- This means we profit from big moves in either direction - not from the stock going up or down specifically
- The hedge is rebalanced daily as the stock moves

Backtest Design Choices

Capital per Trade

Each position uses ~1% of the $1M portfolio - we never bet the farm on one stock

Hold Period

Maximum 3 trading sessions per position - fast in, fast out

Stop-Loss

Cut losses at 5% of the position cost - limits downside on any single trade

Entry Filter

Only enter if the model's prediction exceeds a minimum threshold (0.1%) - avoid noise

Exit Discipline

Exit early if the stock drops out of the top-3 ranking - symmetric entry/exit logic

Post-Announcement Entry

Enter the day after earnings (not before) to avoid buying into inflated option prices

Starting capital: $1,000,000 · Evaluation period: January 2010 – December 2024

Performance Results

	Total Return	Max Loss	Volatility	Trades
Our Strategy	-3.80%	-4.20%	0.52%	124
S&P 500 (SPY)	+579.48%	-33.72%	17.08%	N/A

-$43,687
Total Loss
on $1M capital

28.2%
Win Rate
35 wins / 89 losses

0.41×
Win/Loss Ratio
avg win $524 vs avg loss $1287

-4.20%
Maximum Drawdown
vs -33.7% for S&P 500

What this means: The strategy generated a **-3.80% return with low volatility (0.52%) but poor trade efficiency**. The **maximum drawdown was -4.20%**, significantly smaller than the **S&P 500's -33.7% crisis drawdown**. However, the **win rate was only 28.2% and losses were larger than gains**, resulting in negative profitability despite relatively contained risk.

Risk Controls

-7.16%

Worst Drawdown

Recovered in ~420 days
(S&P 500 lost 33.7%)

0.80×

Win/Loss Ratio

Winners smaller than losers

-0.125%

Tail Risk (VaR)

Worst 1-in-1000 day
is a tiny move

31.5%

Stop-Loss Rate

About half of trades
stopped out - by design

$109K

Max Exposure

Only 10.9% of portfolio
deployed at peak

319

Risk Events

Flagged for review
Zero actual breaches

How We Monitor Risk

Position-Level Controls

- Maximum exposure per position: 1% of capital
- Hard stop-loss at 5% of entry cost
- Maximum hold: 3 trading sessions
- Automatic exit if ranking deteriorates
- Only enter if prediction exceeds threshold

Portfolio-Level Monitoring

- Daily tracking of total portfolio exposure
- Drawdown alerts at -5%, -10%, -15% levels
- Tail-risk measures (VaR and expected shortfall)
- Concentration checks (no single stock dominates)
- Volatility spike detection and response

All risk metrics are computed from actual executed trades - never from hypothetical signals

Honest Assessment

Strengths

•Risk remains well controlled, with a maximum drawdown of -4.20%, far smaller than the ~-33.7% drawdown experienced by the S&P 500 during major crises.

•Low overall volatility, with ~0.52% annualized volatility, indicating the strategy takes relatively limited market risk.

•Strict point-in-time design: the model never accesses future information, eliminating look-ahead bias and ensuring a realistic backtest.

•Dynamic investable universe refreshed annually, helping avoid survivorship bias and stale stock selection.

•Institutional-style risk monitoring, including exposure limits, stop-loss controls, and automated risk-event flagging.

•Clean, modular, and reproducible codebase, making the research framework easy to audit, extend, and deploy.

Limitations

- Negative profitability: the strategy generated -3.80% total return, with -$43,687 realized PnL on $1M capital.
- Low trade quality: the win rate is only 28.2%, and average losses ($1,287) exceed average gains ($524), producing a reward-to-risk ratio of ~0.41×.
- Poor profit efficiency: the profit factor is only ~0.41, meaning losses dominate gains.
- Moderate trading frequency: the strategy executed 124 trades, but trade outcomes remain inconsistent.
- Capital utilization is still limited, with peak exposure of ~$109K (~10.9% of capital).
- Trading costs materially reduce performance, contributing to ~$7.4K in total execution costs.

Should You Invest?

The Case For

- Strong research discipline: strict point-in-time data and no look-ahead bias
- Well-designed risk framework with exposure limits and monitoring infrastructure
- Strategy behaves independently from market direction, making it a potential diversifier
- Modular research pipeline allows easy extension to larger universes and new signals
- Framework is scalable, enabling additional alpha factors to be layered in

The Case Against

- Signal quality is currently weak, with inconsistent trade outcomes
- Capital efficiency is low, as the strategy deploys only a small fraction of available capital
- Concentrated trade structure, with exposure typically limited to a single stock
- Execution costs and short holding windows reduce profitability
- Further research required before institutional deployment

Our Verdict

The project demonstrates a **robust quantitative research framework**, but the **current signal lacks sufficient predictive power** to generate reliable alpha.
The most promising next steps are **improving signal design, expanding the universe, and testing alternative holding horizons** to increase both **trade quality and capital utilization**.

What We'd Improve Next

More Trades

Expand from 30 to 100+ stocks per year - more earnings events = more trades and better diversification

Longer Holding

Test 5, 10, and 20-day holds instead of 3 - give the post-earnings drift more time to play out

Better Signal Testing

Measure prediction accuracy with industry-standard metrics before options are involved

Equity Baseline

Test the predictions with plain stock trades (no options) to isolate how much the model adds vs the options structure

Portfolio Sizing

Scale position sizes based on conviction strength and market volatility instead of fixed 1% per trade

Stress Testing

Run the strategy through the 2008 crisis, COVID crash, and 2022 rate shock scenarios

Full Pipeline at a Glance

1–3

Collect & Clean Data

Quarterly financials (Zacks), daily prices, options data (WRDS), earnings dates (Compustat)

4–5

Build Model Inputs

13 financial + market features, 60-day lookback windows, 124,824 training sequences

6

Train the Model

Expanding walk-forward LSTM, retrained each year, 15 annual folds, 109K OOS predictions

7

Select Universe

Dynamic 30-stock selection from S&P 500, annual rebalance, options-availability filter

8–9

Generate Signals

Earnings-triggered ranking, top-3 selection, option selection (ATM calls, 30–45 DTE)

10–13

Trade & Evaluate

Delta-hedged backtest, P&L engine, performance benchmarks, full risk framework

Key Takeaways

We developed a disciplined machine learning framework to study post-earnings price reactions using strictly point-in-time data and a dynamic stock universe. While the current specification does not produce a reliable trading edge, the results highlight how difficult short-term earnings prediction is and emphasize the importance of signal quality, trade structure, and capital efficiency. The main contribution of this project is the scalable research infrastructure, which provides a strong foundation for further improvements such as expanding the universe, refining features, and testing alternative holding horizons.

1

Company financials predict post-earnings stock moves

2

Machine learning captures patterns humans miss

3

Options let us profit from big moves in either direction

4

Rigorous risk controls keep losses small and recoverable

Next steps: expand universe to 100+ stocks · test longer holding periods · add signal diagnostics · equity-only baseline